

Final Project Report

SMART HOSTEL MANAGEMENT SYSTEM

Course: Object Oriented Programming

Institute: Sukkur IBA University

Department: BS-Software Engineering – Semester II

Instructor: Sir Sher Muhammad Daudpota

Group members

Name	CMS_ID
Falaq Aftab Qazi	053-25-0011
Erum Naz	053-25-0010
Muhammad Tarique	053-25-0031

INTRODUCTION

The Smart Hostel Management System is a desktop-based application designed to automate and improve the day-to-day management of hostel operations. The traditional hostel environments continue to rely on paper-based registers for attendance, billing, and complaint management. This often leads to errors, inefficiency, and lack of proper data organization.

To solve these issues, our project replaces this workflow with a computerized system developed using Java programming language with Object-Oriented Programming (OOP) principles. The system uses Java Swing for Graphical User Interface (GUI) and MySQL database for data storage, connected through JDBC (Java Database Connectivity).

The system is designed with a role-based structure including Admin, Mess In-Charge, and Student, ensuring proper separation of responsibilities and efficient management of hostel operations.

PROBLEM STATEMENT

Hostel management in many institutions is still performed manually or using unorganized digital tools. This creates several problems:

- Difficulty in maintaining accurate student records
- Errors in mess attendance tracking
- Inefficient monthly billing calculation
- Lack of proper communication for complaints
- No centralized system for data management
- Time-consuming manual reporting process

Due to these issues, there is a strong need for an automated system that ensures accuracy, efficiency, and easy access to hostel-related information.

PROJECT GOALS AND OBJECTIVES

The primary goal of the Smart Hostel Management System is to provide a centralized and user-friendly desktop application that manages the day-to-day hostel operations.

Objective	Description
Student Management	Admin can add, update, and delete student records.
Meal Attendance	Mess In-Charge can mark individual or bulk attendance for each meal with the per-meal price from the day's menu.
Menu Management	Mess In-Charge can set daily menus with separate prices for breakfast, lunch, and dinner.
Automated Billing	Monthly bills are auto-calculated from PRESENT attendance records; no manual computation required.
Complaint System	Students can submit complaints; Admin can view and resolve them.
Role-Based Dashboards	Separate Swing dashboards for Admin, Mess In-Charge, and Student roles.

Technical goals:

From a technical perspective, the system is designed to implement role-based access control, ensuring that Admin, Mess In-Charge, and Student users can only access features relevant to their responsibilities. It leverages Object-Oriented Programming principles for modular and scalable design, uses Java Swing for an interactive graphical user interface, and integrates a MySQL database through JDBC for secure and efficient data management.

METHODOLOGY

The system was designed using a **three-tier layered architecture**, ensuring clear separation of concerns and maintainability. Each layer is responsible for a specific part of the application:

Layer	Package	Responsibility
Presentation Layer	gui	Implements Java Swing frames and panels, handles event listeners and user interaction
Business Logic Layer	service	Validates input, applies business rules, and coordinates DAO operations
Data Access Layer	dao + db	Executes JDBC queries on MySQL and maps <code>ResultSet</code> data to model objects
Domain Model Layer	model	Contains plain Java classes representing system entities (User, Student, Billing, etc.)

OOP principles	Implementation
Encapsulation	All model classes use private variables with public getters and setters
Inheritance	Admin, Student, and MessInCharge inherit from the base <code>User</code> class
Abstraction	Service layer hides database and SQL operations from the GUI layer
Polymorphism	<code>getDescription()</code> is overridden; <code>calculateBill()</code> is overloaded in <code>BillingService</code>

Development Process

The system was developed using an **iterative approach**, where each module was built and tested step by step:

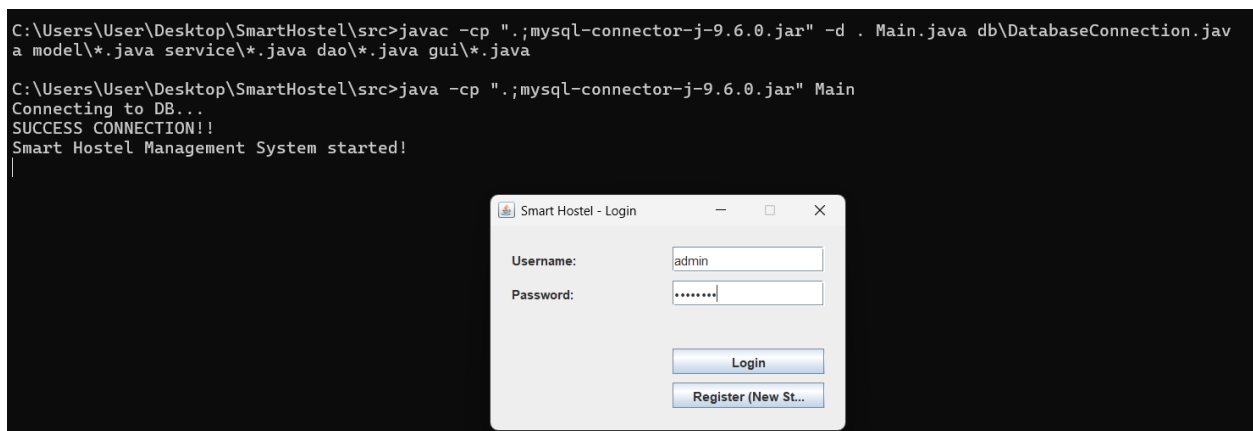
- The database schema was designed first (`database_setup.sql`), including tables for users, students, attendance, menu, billing, and complaints.
- Model classes were created for each entity, establishing the object-oriented structure and relationships.
- DAO classes were implemented using parameterized JDBC queries to ensure secure database interaction and prevent SQL injection.
- Service classes were developed to handle validation, business logic, and coordination between GUI and database layers.
- The graphical user interface (GUI) was built using Java Swing, connecting user actions to backend services.
- Finally, end-to-end testing was performed using sample login credentials.

Project Implementation

The system was implemented as a Java Swing desktop application with role-based dashboards. Each user (Admin, Mess In-Charge, Student) interacts with the system through dedicated interfaces connected to a MySQL database.

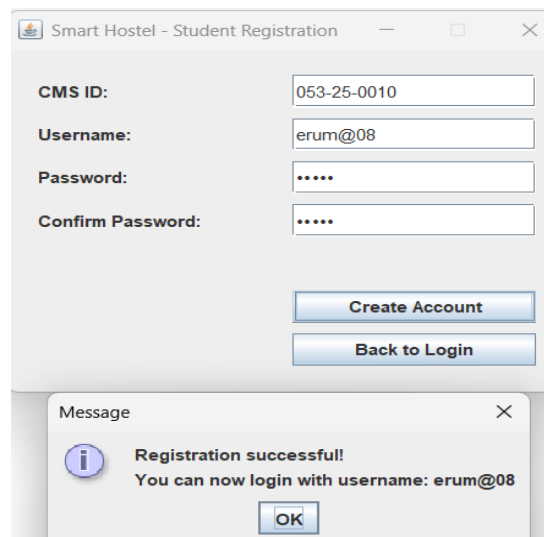
Login Screen

The login module authenticates users using username and password and redirects them to their respective dashboards based on assigned roles (Admin, Student, Mess In-Charge).



Registration

The registration module allows new users to enter their details, which are validated and securely stored in the system database.



Admin Dashboard

Tab 1 — Student Management

This module enables the admin to add, update, and delete student records efficiently.

The screenshot shows the 'Admin Dashboard - admin' window with the 'Manage Students' tab selected. It features a 'Student Information' form with fields for CMS ID, Name, Room No, Department, and Contact. Below the form is a table of student records.

ID	CMS	Name	Room	Department	Contact
1	053-25-0011	Falaq Aftab Qazi	207	BS Software engineering	03245320714

Tab 2 — Complaint Management

This module allows the admin to view all student complaints and update their status to “Resolved”.

The screenshot shows the 'Admin Dashboard - admin' window with the 'View Complaints' tab selected. It displays a table of student complaints. A modal message box is open, indicating that a complaint has been marked as resolved.

#	CMS ID	Student Name	Complaint	Status	Date
1	053-25-0011	Falaq Aftab Qazi	Wifi is Not Working Properly on 2nd Floor, Please Resolve this iss	RESOLVED	2026-05-03 03:01:58

Message
Complaint marked as resolved!
OK

Mark as Resolved Refresh

Mess Dashboard (Mess In-Charge)

Tab 1 — Meal Attendance

This module allows marking meal attendance for students either individually or in bulk.

Mess Dashboard - mess

Mess In-Charge Dashboard Logout

Mark Attendance Manage Menu Generate Bill

Mark Attendance for: 2026-05-03

Meal: DINNER Status: PRESENT Load Students Mark Selected Mark All Present

CMS ID	Name	Room	Status
053-25-0011	Falaq Aftab Qazi	207	PRESENT

Message

Attendance marked: PRESENT for DINNER
Student: Falaq Aftab Qazi
Price recorded: PKR 200.0

OK

Tab 2 — Menu Management

This module allows setting the daily menu along with pricing for each meal.

Mess Dashboard - mess

Mess In-Charge Dashboard Logout

Mark Attendance Manage Menu Generate Bill

Set Daily Menu

Date (YYYY-MM-DD): 2026-05-03

Breakfast: Aloo Paratha, Tea

Breakfast Price (PKR): 80

Lunch: Chicken Biryani

Lunch Price (PKR): 150

Dinner: Mutton Pulao

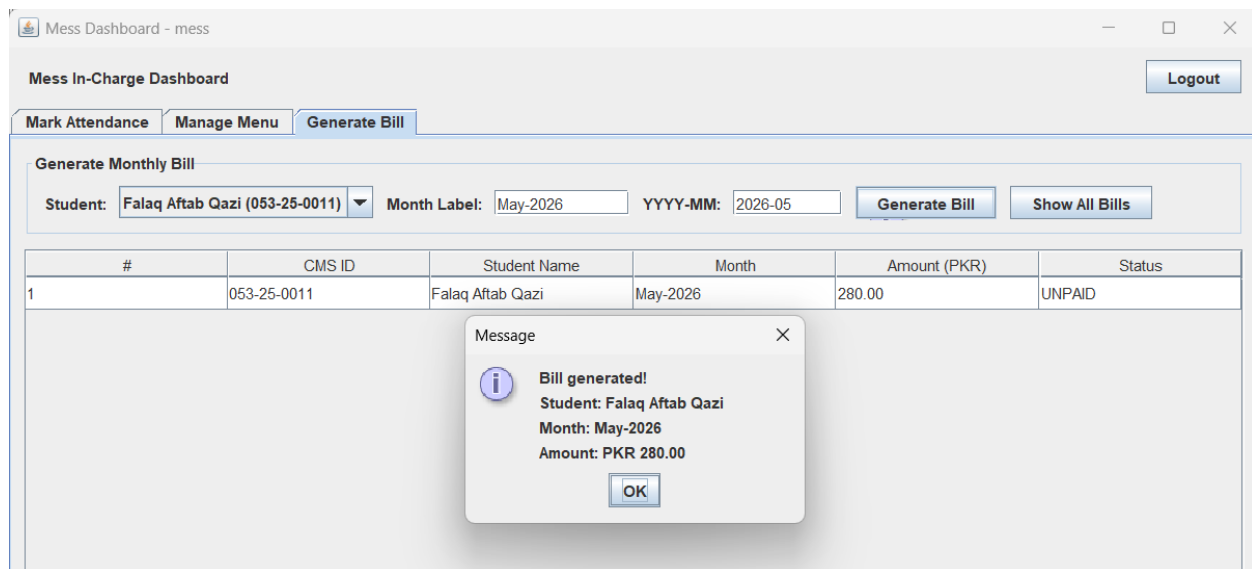
Dinner Price (PKR): 200

Save Menu Load Today's Menu

Prices are automatically used when marking attendance.

Tab 3 — Billing System

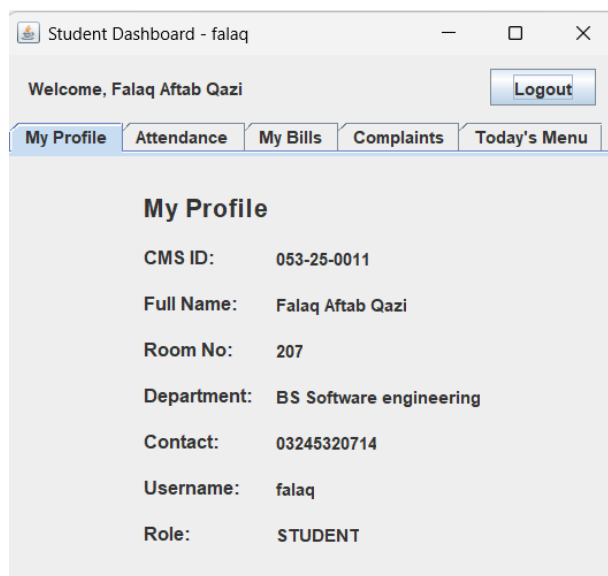
This module generates and displays monthly bills based on student meal attendance and menu pricing.



Student Dashboard

Tab 1 — My Profile

This module displays the student's personal and hostel-related information.



Tab 2 — Attendance

This module allows students to view their meal attendance records.

 Student Dashboard - falaq

Welcome, Falaq Aftab Qazi

Logout

My Profile

Attendance

My Bills

Complaints

Today's Menu

Refresh Attendance

CMS ID	Name	Date	Meal	Status	Price (PKR)
053-25-0011	Falaq Aftab Qazi	2026-05-03	BREAKFAST	PRESENT	80.00
053-25-0011	Falaq Aftab Qazi	2026-05-03	DINNER	PRESENT	200.00

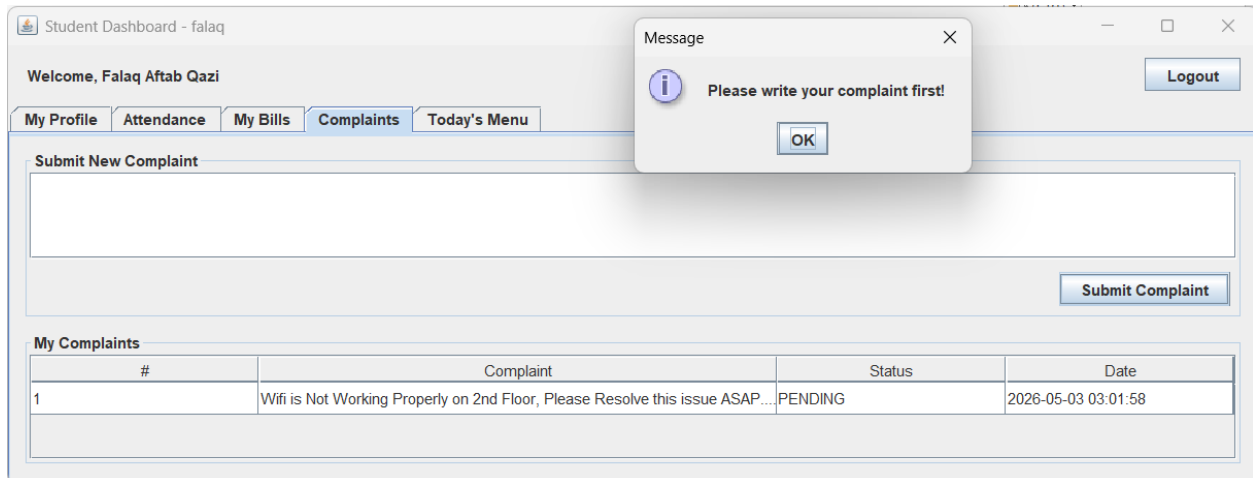
Tab 3 — My Bills

This module enables students to view their generated monthly mess bills.

Student Dashboard - falaq			
Welcome, Falaq Aftab Qazi			Logout
My Profile	Attendance	My Bills	Complaints Today's Menu
Refresh Bills			
#	Month	Amount (PKR)	Status
1	May-2026	280.00	UNPAID

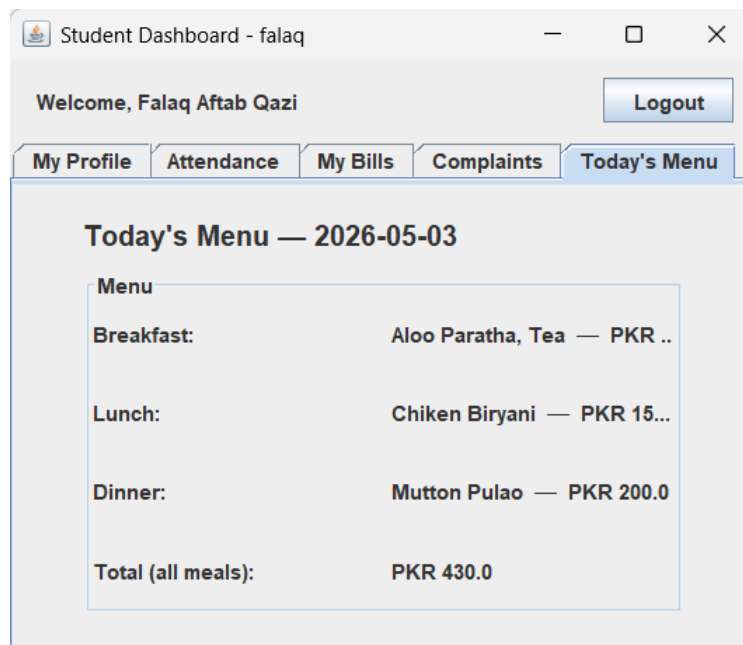
Tab 4 — Complaints

This module allows students to submit and track complaints.



Tab 5 — Today's Menu

This module displays the daily menu along with meal pricing.



SYSTEM WORKING FLOW

1. User logs into the system
2. System verifies role from database
3. User is redirected to respective dashboard
4. User performs operations (attendance, student entry, etc.)
5. Data is stored/retrieved from MySQL database
6. Results are displayed in GUI tables/forms

MAJOR OUTCOMES

Following Major outcomes proposed in the idea submission were achieved:

- 1.Role-based login routing Admin / Mess In-Charge / Student to separate dashboards
2. Admin can perform full CRUD on student records
3. Mess In-Charge can set daily menu with per-meal prices
4. Meal attendance marked individually or in bulk; price auto-fetched from menu
5. Monthly billing auto-calculated from PRESENT attendance records
- 6.Student can view profile, attendance history, bills, and today's menu
- 7.Student complaint submission with Admin resolution workflow
- 8.All four OOP pillars demonstrated in the codebase

PROJECT TIMELINE

Phase	Title	Key Activities	Duration
1	Planning	Idea selection, problem definition, role assignment, proposal writing	3 days
2	Idea Submission	Finalized proposal and submitted 5-minute idea video	3 days
3	Database Design	Designed ERD and created/tested database setup	3 days
4	Backend Coding	Developed Model, DAO, Service classes and tested JDBC	6 days
5	GUI Development	Built Swing dashboards for Admin, Student, and Mess roles	6 days
6	Testing & Report	Testing, bug fixing, report writing, and demo video creation	3 days

CONCLUSION

The Smart Hostel Management System provides an efficient and automated solution for managing hostel operations. By integrating Java OOP concepts, Swing-based GUI, and MySQL database connectivity, the system ensures structured data management and reduces manual workload.

The role-based design improves system organization by separating responsibilities among Admin, Mess In-Charge, and Students. This project also enhances practical understanding of software development, database management, and object-oriented programming.

The video presentation for this project is available at the link provided below:

https://drive.google.com/file/d/1eEY-SWN6h3cBJgi_OVaAck8SzZqnj5BE/view?usp=drivesdk

https://drive.google.com/file/d/1eEY-SWN6h3cBJgi_OVaAck8SzZqnj5BE/view?usp=drive_link